Dense Layer

April 23, 2026 7:09 PM

Forward pass:

$$\underset{\text{input}}{\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}} \begin{bmatrix} \overset{\text{Neuron 1}}{w_{11}} & & \overset{\text{Neuron n}}{w_{1n}} \\ w_{21} & \cdots & w_{2n} \\ w_{31} & & w_{3n} \end{bmatrix} + \underset{\text{bias}}{\begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix}} = \underset{\text{output}}{\begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix}}$$

Backward pass:

we need to calculate dL/dw. we are given dL/dy, so we need dy/dw

$dy/dw \rightarrow y = w_1 x_1 \ldots + w_n x_n + \text{bias}$

then $dy/dx_1 = x$, then dy/dx (ALL) is just X

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} dL/dy_1 & \cdots & dL/dy_n \end{bmatrix} = \begin{bmatrix} dL/dw_{11} & & dL/dw_{1n} \\ \vdots & \cdots & \vdots \\ dL/dw_{n1} & & dL/dw_{nn} \end{bmatrix}$$

The gradient of each weight

Passing gradients to previous layer:

we want to return dL/dx. we have dL/dy

so we need to calculate $dL/dy \cdot dy/dx$

which can be done with:

$dL/dy \cdot w^T$ → same chain rule as backward pass, applied to a different variable